Jetson board introduction

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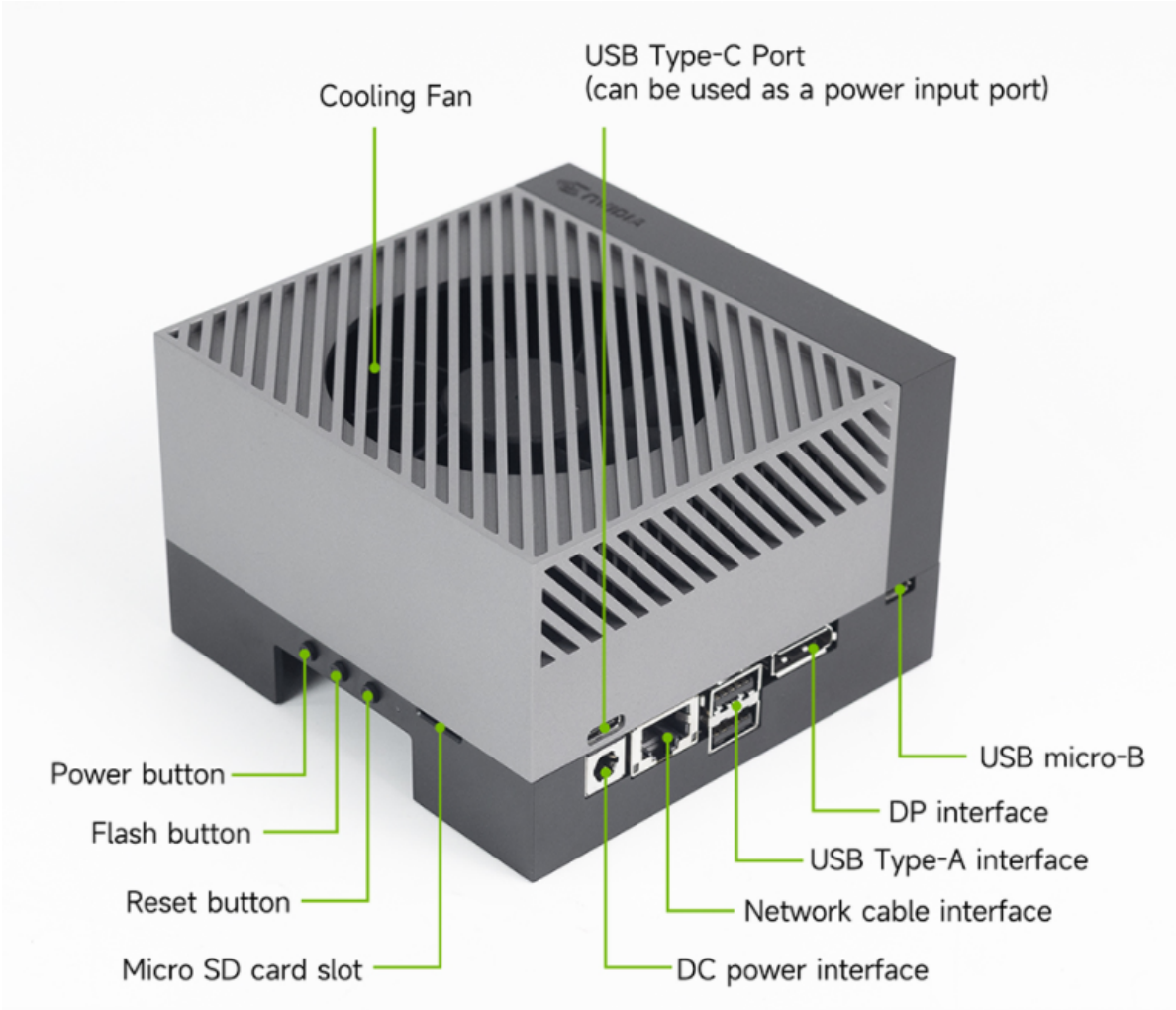
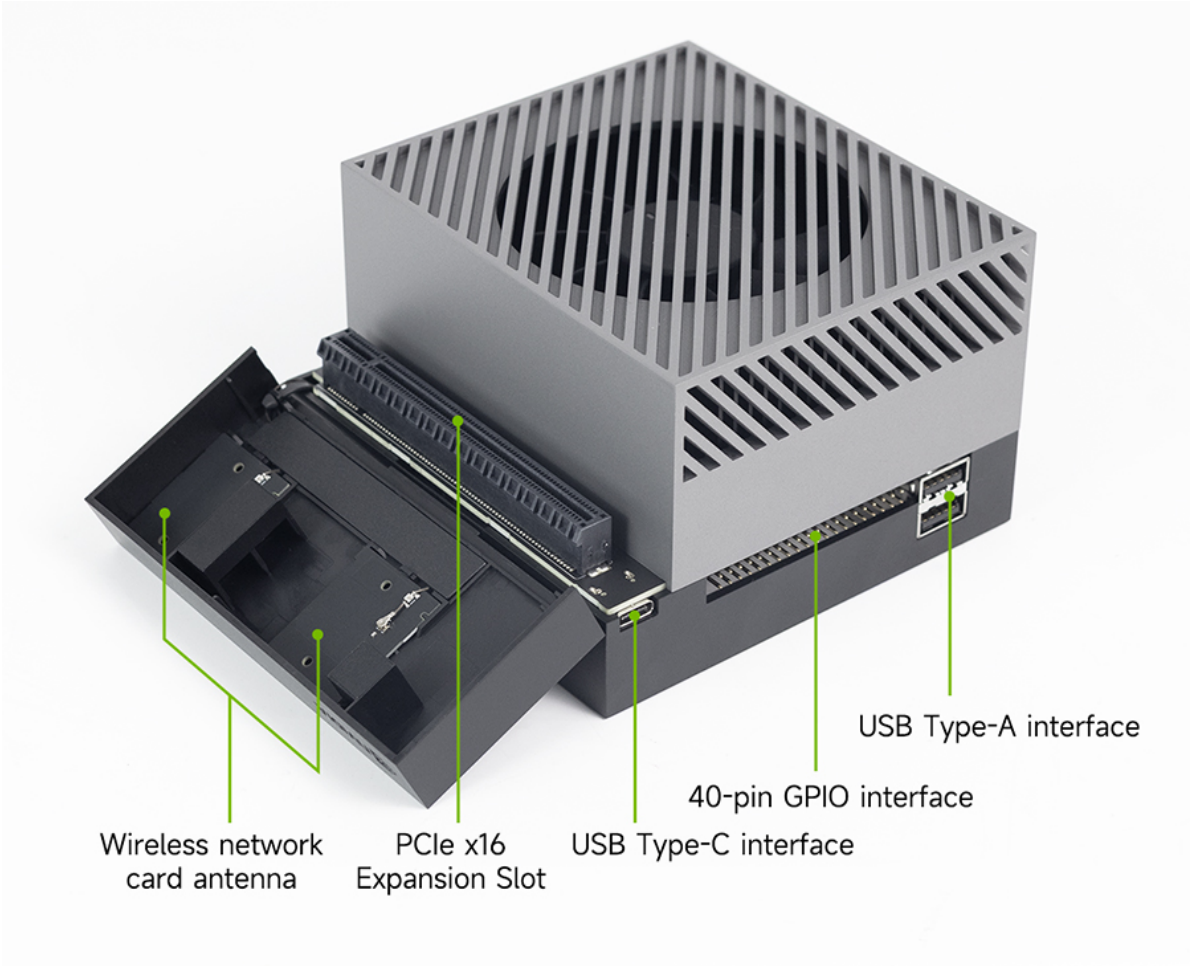
1. Jetson AGX Orin Development Kit
2. Jetson AGX Orin Module Parameters

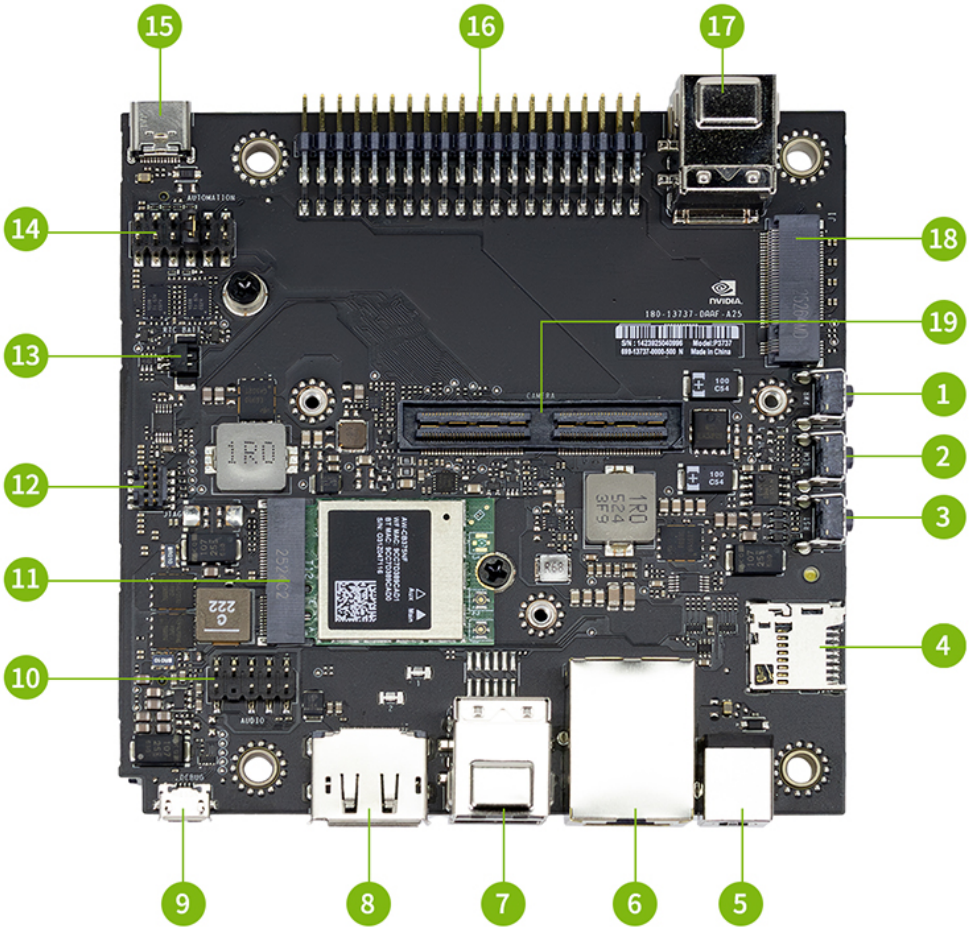
1. Jetson AGX Orin Development Kit



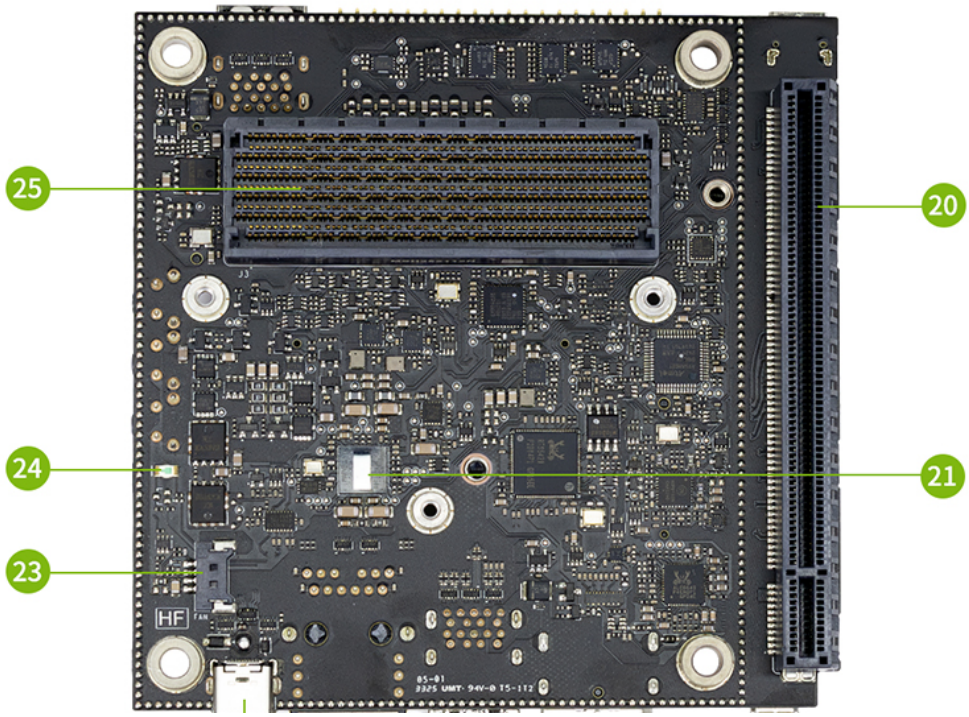

USB Type-A interface
40-pin GPIO interface
Wireless network card antenna
PCIe x16 Expansion Slot
USB Type-C interface

Carrier Board Functional Distribution

Front



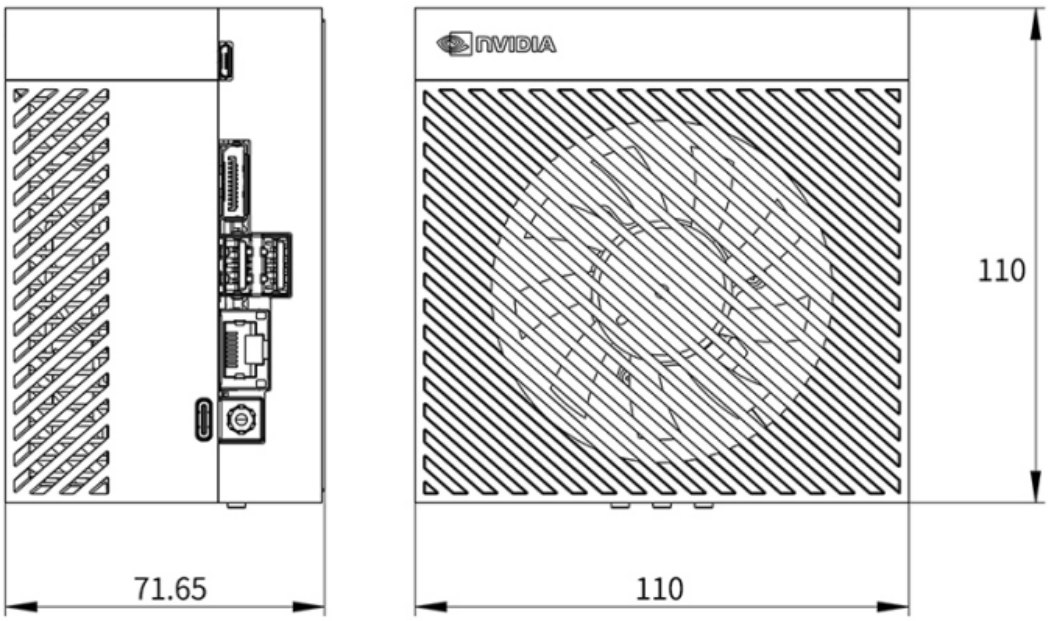
Back



① Power button
② Force recovery button (flashing button)
③ Reset button
④ microSD card slot
⑤ DC power connector
⑥ RJ45 Ethernet port
⑦ Two USB Type-A ports
⑧ Display output port
⑨ USB Micro-B port
⑩ HD Audio connector
⑪ M.2 E-Key slot (for wireless network card)
⑫ JTAG debugging port
⑬ RTC spare battery connector
⑭ Automation control interface
⑮ USB Type-C interface
⑯ 40-pin expansion interface (GPIO expansion header)
⑰ Two USB Type-A interfaces
⑱ M.2 M-Key slot (for NVMe SSDs)
⑲ Camera interface
⑳ PCIe x16 expansion slot (for graphics cards/capture cards, etc.)
㉑ 10 Gigabit Ethernet chip
㉒ USB Type-C interface
㉓ Fan connector
㉔ White LED indicator
㉕ Jetson core module interface

Dimensions

Unit: mm



2. Jetson AGX Orin Module Parameters

Table 3: Jetson AGX Orin Developer Kit Technical Specifications

MODULE:	
GPU	NVIDIA Ampere architecture with 2048 NVIDIA® CUDA® cores and 64 Tensor cores
CPU	12-core Arm Cortex-A78AE v8.2 64-bit CPU 3MB L2 + 6MB L3
DL Accelerator	2x NVDLA v2.0
Vision Accelerator	PVA v2.0
Memory	32GB 256-bit LPDDR5 204.8 GB/s
Storage	64GB eMMC 5.1
Power	15W to 60W

REFERENCE CARRIER BOARD:	
Camera	16 lane MIPI CSI-2 connector
PCIe	x16 PCIe slot supporting x8 PCIe Gen4
M.2 Key M	x4 PCIe Gen 4
M.2 Key E	x1 PCIe Gen 4, USB 2.0, UART, I2S
USB	Type C: 2x USB 3.2 Gen2 with USB-PD support Type A: 2x USB 3.2 Gen2, 2x USB 3.2 Gen1 Micro-B: USB 2.0
Networking	RJ45 (up to 10 GbE)
Display	DisplayPort 1.4a (+MST)
microSD slot	UHS-1 cards up to SDR104 mode
Others	40-pin header (I2C, GPIO, SPI, CAN, I2S, UART, DMIC) 12-pin automation header 10-pin audio panel header 10-pin JTAG header 4-pin fan header 2-pin RTC battery backup connector DC power jack Power, Force Recovery, and Reset buttons
Dimensions	110mm x 110mm x 71.65mm (Height includes feet, carrier board, module, and thermal solution)